

**A Web-Based Ordering and Automated Inventory Management
System for MRJE Gas and Bright Star
Water Refilling Station**

A Software Design Project

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

E-commerce, or electronic commerce, is the digital buying of goods and services between businesses and their consumers (B2B). It offers a wider selection of goods without requiring consumers to visit a physical store. In today's digital age, online shopping has become an important part of business, changing how companies and customers connect with each other. It enables faster transactions and efficient delivery of products and services. The technological revolution today has made the use of e-commerce platforms an essential tool for business growth, allowing businesses to compete in the digital marketplace.

The implementation of e-commerce platforms provides improved operational management of business transactions and offers convenience for customers to easily select their desired products and services, giving a more engaging and satisfying shopping experience (Qurtubi et al., 2024; Zulkarnain, 2023). This study highlights that adoption of e-commerce with digital transformation provides multiple advantages to improve logistics and marketing operations to optimize internal processes, improve supply chains, and be able to meet market demands (Jain et al., 2021; Sharma et al., 2023). Moreover, the adoption of e-commerce in business operations also supports market competition and the continuous emergence of business strategies.

However, the implementation of e-commerce as part of digital transformation in business transaction settings presents challenges. These challenges include data security and privacy, regulatory compliance, and market competition (Plekhanov et al., 2023). Addressing these challenges and identifying appropriate solutions offers sustainable business growth and fosters customer loyalty, which enhances the competitiveness and enables businesses to adapt effectively to ongoing digital evolution.

This study aims to examine the role of e-commerce in contemporary business, its impact on organizational efficiency and market reach, and the opportunities and challenges presented by digital transformation. By combining ideas from recent studies, the study aims to clearly explain how businesses can use digital technology to succeed in today's digital market.

Conceptual Model

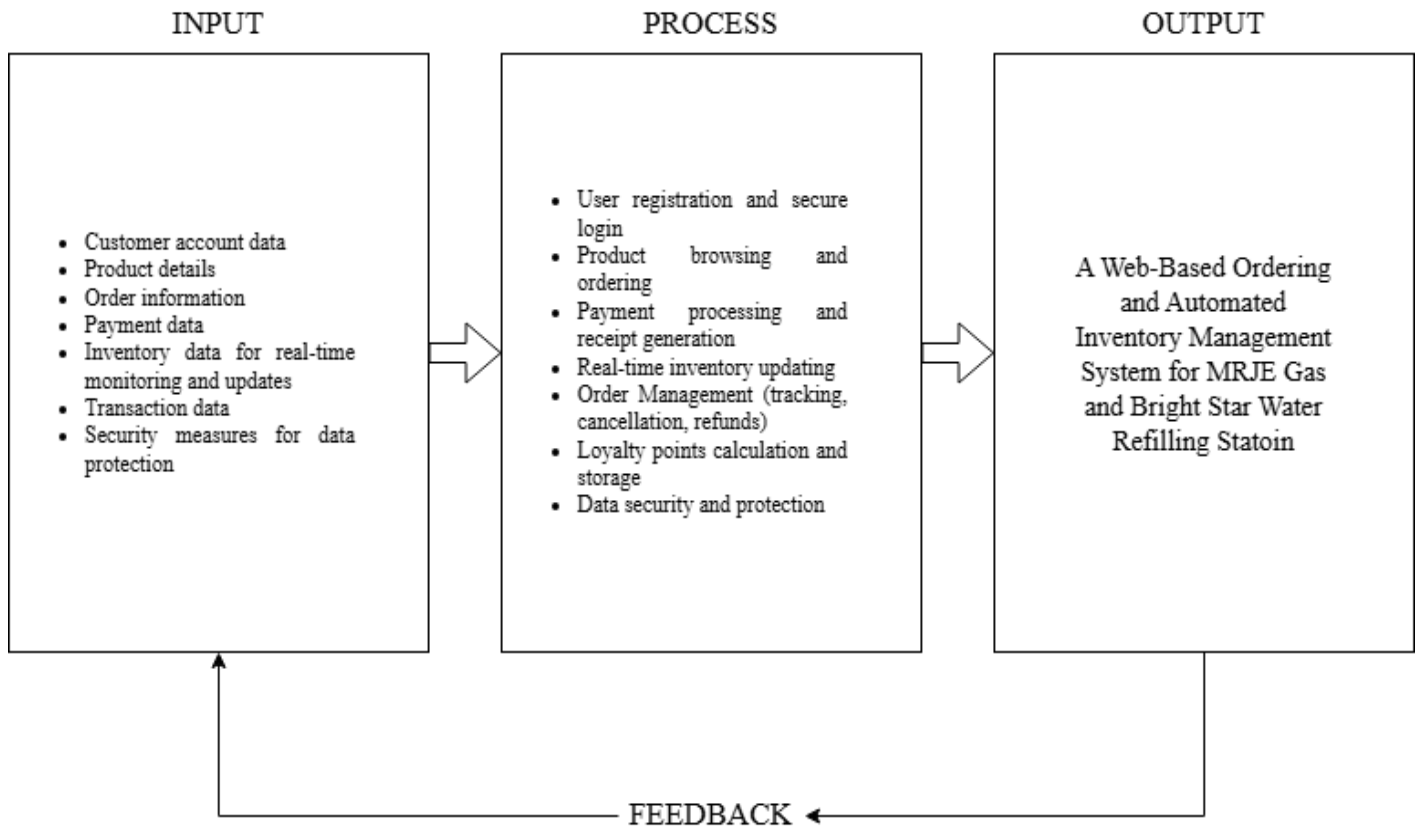


Figure 1. *The Conceptual Model of the Study.*

Figure presents the Input–Process–Output (IPO) model of the study. The input stage includes essential data required for system operation, such as customer account information, product details, order information, payment data, inventory records for real-time monitoring, transaction data, and security measures for data protection. These inputs are then processed through key system functions, including user registration and secure login, product browsing and ordering, payment processing with receipt generation, real-time inventory

updating, and order management features such as tracking, cancellation, and refunds. The process also incorporates loyalty points calculation and storage, along with the implementation of data security protocols to ensure safe and reliable transactions. As a result, the system produces an efficient and fully functional web-based ordering and automated inventory management platform, designed to streamline business operations, improve transaction accuracy, and enhance customer experience for MRJE Gas and Bright Star Water Refilling Station.

Objectives of the Study

The Web-Based Ordering and Inventory Management System aims to improve the efficiency of ordering, inventory monitoring, and customer transactions through an online platform. The study had the following specific objectives:

1. A system that allows customers to register, log in, and manage their accounts with secure authentication and encrypted credentials.
2. A system that allows customers to view available products, place orders, and schedule deliveries.
3. A system that supports payment through Cash on Delivery (COD) and GCash with digital receipt generation and secure transactions.
4. A system that allows staff to monitor and update inventory in real time.

5. A system that enables delivery tracking, order cancellation, refunds, and loyalty points management with secure data protection.

Scope and Delimitation

This study covers the development of a Web-Based Ordering and Inventory Management System designed to improve the efficiency, accuracy, and convenience of business operations for MRJE Gas and Bright Star Water Refilling Station. The system allows customers to access the platform using internet-connected devices such as computers, tablets, and smartphones. It provides key functionalities including online ordering, delivery scheduling, digital receipt generation, and secure user authentication with encryption to protect customer data and transactions. It also includes inventory management and order processing features, enabling authorized personnel to monitor stock levels, manage orders, and maintain customer records. The system supports payment methods such as GCash via QR code and Cash on Delivery (COD), offering flexible payment options for users.

The system also incorporates a loyalty rewards feature that provides cashback incentives through a point-based mechanism. Customers earn one loyalty point for every ₱100 spent, which is equivalent to approximately 1% cashback, provided that each transaction meets the minimum purchase requirement of ₱500. Furthermore, customers who complete at least three qualified orders, each worth ₱500 or more, within a two-week period are granted

additional bonus points. This increases the effective cashback rate to approximately 2%. The total loyalty points earned are computed by combining base and bonus points, and these points are directly convertible into monetary value at a fixed rate of one point equivalent to one peso.

The system operates within a defined service area covering an approximate 10-kilometer delivery radius from Brgy. San Lorenzo Ruiz, San Pedro, Laguna. It utilizes an estimated distance-based model to compute delivery fees, where the minimum distance of 0 to 3 kilometers from the store location is eligible for free delivery. For distances beyond 3 kilometers up to 6 kilometers, a fixed delivery fee of ₱30 is applied, while distances greater than 6 kilometers up to 10 kilometers are charged ₱50. These delivery fees are determined based on predefined distance zones rather than real-time GPS tracking. The system includes products offered by the station, such as MRJE LPG products and accessories, branded LPG items, and purified drinking water products including 5-gallon containers and bottled water.

However, the system is limited to a single-store operation and does not support multi-branch management or scalability for larger business networks. It requires a stable internet connection and cannot function in offline environments. Payment options are limited to GCash and Cash-on-Delivery (COD), excluding other methods such as credit or debit cards. Delivery management is restricted to scheduling and status updates only, without real-

time GPS tracking or route monitoring, and delivery confirmation is handled by the administrator since there is no driver-side application. Additionally, the system does not include a dedicated mobile application for delivery personnel, as all operations are conducted through the web interface. Inventory and order monitoring rely on manual updates by authorized staff, and the system does not implement advanced automation or real-time tracking technologies. Integration with third-party logistics services is also beyond the scope of this study. The loyalty rewards system is limited to a fixed computation model and does not include variable rates, tiered memberships, or promotional adjustments beyond the defined cashback and bonus conditions.

Significance of the Study

This study contributed to demonstrating how a web-based ordering and inventory management system with a customer database and loyalty features can improve business operations and customer experience. The research result would benefit the following:

Business Owners. The system may help improve inventory monitoring, order processing, and delivery scheduling through ensuring accurate inventory and automated transactions. It may simplify the payment and support better control over sales and customer data management.

Customers. The system may provide convenience by providing faster transitions and easy access to purchase history. It may also offer incentives through loyalty rewards which may encourage repeat purchases and enhance customer satisfaction.

Staff and Delivery Personnel. The system may reduce workload by automating inventory management, order tracking, delivery updates, and processing cancellations and refunds, helping staff to provide reliable service and manage operations smoothly.

Organizations Adopting E-Commerce. The study may serve as a reference for small to medium-sized businesses planning to adopt web-based ordering systems and inventory systems. It demonstrates the practical use of digital receipts, e-wallet payments, loyalty rewards, and automated refund for improved business transactions.

Academe. Trimex Colleges, Inc., particularly the Computer Engineering and IT/Computer Studies programs, may use the system as a teaching reference for web-based system design, database management, and e-commerce application development. It can help instructors and students gain practical experience in digital business operations.

Software Developers. The system may guide professionals in building and designing efficient web-based applications with loyalty features, inventory control, customer databases, and payment processing, supporting practical e-commerce applications.

Future Researchers. This study may serve as a learning resource for students and researchers interested in building a similar e-commerce system and exploring opportunities to improve functionalities and create more advanced features.

Definition of Terms

This section outlines essential terminology relevant to the design project. The provided definitions aim to establish a clear understanding of the technical, operational, and user-related aspects that support the system's development and implementation.

Authentication. A security process that verifies the identity of users before granting access to the system, typically through login credentials such as usernames and passwords.

Consumers. Individuals who use the system to browse products, place orders, and complete transactions for personal use through the web-based platform.

Cash-On-Deliver (COD). A payment method where customers pay for their orders in cash upon delivery of the goods.

Encryption. A data protection method that converts sensitive information into a secure format to prevent unauthorized access, ensuring the confidentiality of user data and transactions.

E-commerce. This refers to the process of buying and selling goods or services through electronic platforms, particularly web-based systems, where transactions, payments, and order processing are conducted online.

GCash. A mobile wallet and online payment service in the Philippines that allows users to send and receive money, pay bills, and complete digital transactions using a smartphone.

Inventory System. A computerized mechanism used to monitor, manage, and control product stock levels, including tracking item availability, quantities, and updates resulting from sales and restocking activities.

Web-based Ordering System. A digital platform or application that enables users to place orders for products or services over the internet. This system typically allows customers to browse a catalog of available items, select products, customize their orders if needed, and complete transactions through secure payment gateways.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter will present relevant literature and studies gathered from research articles, journals, case studies, and online sources written by foreign and local scholars, which will serve as the foundation for the development of the proposed system. It will highlight existing systems, technologies, and methodologies related to web-based ordering, inventory management, customer databases, and digital payment systems, while also examining security practices such as user authentication and data encryption to ensure the protection and confidentiality of user information. Furthermore, this chapter will identify similarities and differences between previous studies and the current research and will discuss how the proposed system will address gaps found in earlier works through the integration of secure, efficient, and reliable system features.

Foreign Literature

An e-commerce system, also referred to as a web-based ordering, inventory, and customer management system, has become an essential tool for modern business operations and marketing, as it helps increase sales and enhance customer engagement. These systems integrate key features such as stock monitoring, customer databases, and streamlined transactions, which reduce human error and support better decision-making (Liew & Nayef, 2022). In addition, such systems are particularly beneficial for small and medium-sized enterprises because they enable automation of order processing, accurate

record-keeping, and improved supply chain management. Furthermore, the integration of analytical tools allows businesses to forecast demand trends, thereby improving operational efficiency and strategic planning (Liew & Nayef, 2022).

Building on these system capabilities, consumer behavior and digital marketing strategies significantly influence the effectiveness of e-commerce platforms. User-friendly interfaces, personalized product recommendations, and promotional offers greatly affect consumers' online purchasing decisions (Hodges, Lowery, Patel, McInnis, & Zhang, 2023). In relation to this, social media engagement and digital promotions contribute to increased sales and stronger customer loyalty over time (Rosário & Raimundo, 2021). These findings emphasize that understanding consumer preferences and continuously adapting digital experiences are essential for sustaining competitiveness in online markets.

In connection with consumer engagement, the implementation of digital loyalty programs has emerged as a key strategy for enhancing customer retention and encouraging repeat purchases. When loyalty programs are aligned with customer satisfaction, they create perceived value, strengthen engagement, and foster emotional connections with the brand (Panjaitan, 2021). Similarly, loyalty programs enhance brand commitment, increase customer retention, and contribute to long-term customer value on digital platforms (Primasari & Dwita,

2025). Thus, web-based systems that incorporate reward mechanisms, interactive engagement tools, and exclusive incentives are more likely to achieve higher customer satisfaction and improved business performance.

Despite these advantages, concerns regarding privacy and security remain critical issues in e-commerce systems. Consumers are becoming increasingly aware of the risks associated with online transactions and data collection, highlighting the need for strong data protection policies (Yao & Tarofder, 2024). In addition, vulnerabilities such as hacking, data breaches, and fraud can significantly damage customer trust and brand reputation (Sagor & Natarajan, 2022). To address these challenges, the implementation of security measures such as authentication, encryption, multi-factor verification, and real-time monitoring is essential for protecting financial data and maintaining consumer confidence (Judichael & Jiang, 2022).

Moreover, the efficiency of e-commerce systems is closely linked to effective business process management (BPM). A well-structured BPM framework improves order processing, inventory control, and customer service operations (Nathaniela et al., 2022). In addition, analyzing user behavior, including browsing patterns and decision-making processes, enables the development of more intuitive, accessible, and user-friendly platforms (Wahyu et al., 2025). These findings indicate that both process optimization and user-centered design are essential for enhancing overall system performance.

In terms of operational strategies, logistics and technological advancements are vital in influencing consumer decisions and overall business success. Offering incentives such as free shipping increases conversion rates and encourages higher purchase volumes (Caracara-Flores, Carnero Quispe, Delgado-Paz, & Pericón-Balderrama, 2025). Similarly, e-commerce expands market reach and strengthens business resilience in the modern economy (Guerrero Vidal, Miranda Cansing, & Chávez Arizala, 2025). Supporting these findings, technological frameworks such as mobile integration, cross-platform compatibility, and secure payment gateways enhance transaction efficiency and improve user experience (Sulova, 2023).

Finally, legal and regulatory considerations are fundamental to the safe and sustainable operation of e-commerce systems. Compliance with data protection laws, consumer rights, and taxation policies ensures secure and lawful transactions while safeguarding both businesses and consumers (Sanger & Maharishi, 2022). As digital commerce continues to evolve, emerging issues such as cross-border taxation and digital consumer rights highlight the need for continuous updates in legal frameworks to keep pace with technological advancements.

Local Literature

The rapid advancement of information technology has significantly transformed business operations, particularly in inventory management and sales processing systems, prompting organizations to adopt computerized and web-based solutions to improve efficiency, accuracy, and customer engagement. In line with this development, existing literature emphasizes the importance of automated inventory monitoring and database management in enhancing operational performance. This also aims similarly to our project to enable real-time monitoring of products while maintaining accurate digital records. Unlike earlier systems that primarily focused on internal inventory processes, the proposed system extends its functionality by incorporating customer ordering features and loyalty rewards to strengthen customer retention. Supporting this, literature indicates that reward-based mechanisms such as discount points and promotions significantly enhance customer satisfaction and encourage repeat purchases (Barrera et al., 2022).

Furthermore, consumer adoption of e-commerce platforms is largely influenced by perceived usefulness, ease of use, and reliability, suggesting that a well-designed web-based system can effectively cater to both new and existing users (Romero et al., 2021). This is further reinforced by findings that efficient operational practices contribute to improved financial performance in LPG retail businesses, highlighting the importance of integrating ordering and inventory systems (Tanaman et al., 2023). In addition, the rapid growth of e-commerce and

digital payment systems in the Philippines highlights how secure, user-friendly platforms contribute to increased consumer spending and business efficiency, particularly when supported by reliable infrastructures and policies (Anne, 2024).

For technological advancements, challenges associated with manual processes continue to justify the need for system automation. Manual inventory systems are often characterized by slow transaction processing, inaccurate stock records, and reporting errors, which hinder overall efficiency and decision-making (Maglinte et al., 2025). Addressing these issues, the proposed web-based system enhances convenience by allowing customers to place orders online, thereby reducing in-store congestion and improving service efficiency. Moreover, studies in the Philippine context reveal that web-based platforms and mobile technologies significantly influence purchasing behavior by providing accessible product information and convenient transaction options (panjaitan et al., n.d.), reinforcing the relevance of integrating an online ordering feature into the system. Furthermore, the increasing adoption of digital payment platforms such as GCash demonstrates how secure and accessible payment systems support the growth of e-commerce by enabling fast, cashless transactions, although concerns regarding trust, cybersecurity, and financial literacy still affect adoption among some users (Anne, 2024). This highlights the importance of incorporating secure payment gateways into the proposed system.

Business performance is also shaped by financial management and consumer behavior. Literature highlights that financial performance is significantly influenced by effective internal management practices, cost efficiency, and sales strategies, which are essential for small and medium enterprises to remain competitive in local markets (Polinar et al., 2024). This suggests that traditional retail businesses, such as retail stores, require not only efficient inventory systems but also integrated solutions that support better financial control and sales monitoring. In relation to customer engagement, satisfaction in e-commerce environments is largely determined by how well online platforms meet consumer expectations in terms of service quality, delivery efficiency, and product reliability (Rufano & Gervacio, 2025). When expectations are not met, customer dissatisfaction may arise, negatively affecting repeat purchase intentions. Supporting this, the expansion of digital payment usage in the Philippines has been shown to increase consumer spending and improve business sales performance, particularly when systems are perceived as secure and easy to use (Anne, 2024), further emphasizing the importance of integrating trusted payment solutions into the proposed system.

Supporting this, factors such as sales promotions, retailer reputation, and user experience have been identified as key determinants of consumer behavior in e-commerce platforms (Yong et al., 2023). These factors positively influence purchasing decisions, as customers are more likely to engage with platforms that offer attractive promotions, trustworthy services, and seamless navigation. This highlights the importance of integrating a customer database and loyalty feature

in the proposed system, allowing businesses to track purchasing behavior, reward loyal customers, and develop targeted marketing strategies. Additionally, the effectiveness of computerized inventory systems has been widely recognized in improving transaction speed, record accuracy, and overall productivity through real-time tracking and automated reporting (Buhia, n.d.). Complementing this, research in food service delivery shows that customer satisfaction is strongly influenced by the efficiency, accessibility, and convenience of online ordering systems (Abay-abay et al., 2021), further supporting the integration of such features to enhance user experience and encourage repeat patronage. Moreover, the continued expansion of e-commerce platforms in the Philippines, supported by secure and efficient digital payment systems, reinforces the role of integrated technologies in enhancing both operational efficiency and customer satisfaction (Anne, 2024).

As well as the increasing adoption of digital payment systems strengthens the relevance of web-based platforms in modern retail environments. The success of information systems in mobile commerce significantly affects consumer behavior, as users tend to adopt platforms that are efficient, reliable, and easy to use (Pelegrin, 2021). In the Philippine setting, the growing preference for cashless transactions is driven by factors such as ease of use, trust, security, and social influence, with consumers favoring businesses that offer digital payment options (Jacob, 2024). These findings are supported by evidence that perceived security and ease of use are critical factors influencing the adoption of digital payment systems, highlighting the need for platforms to implement robust cybersecurity

measures and user-friendly interfaces (Anne, 2024). These findings highlight the importance of incorporating secure and user-friendly cashless payment features, such as GCash integration, into the proposed system to meet customer expectations and improve operational efficiency. Furthermore, integrating customer databases with loyalty mechanisms enables businesses to leverage consumer behavior patterns, thereby enhancing engagement and long-term retention, which aligns directly with the objectives of the study.

Foreign Studies

Several studies have examined the impact of e-commerce on business growth, consumer engagement, and economic development. Digital platforms have been shown to empower local communities by connecting consumers with local producers, thereby fostering sustainable consumption and supporting small-scale enterprises (Lucas, 2023). In a related context, improvements in e-commerce database systems during the COVID-19 pandemic enhanced data management, streamlined operational processes, and improved customer interaction in online marketplaces (Ahmed, Sreejith, & Abdullah, 2021). These developments are further supported by findings that highlight the importance of loyalty programs, user satisfaction, and trust in influencing repeat purchase behavior in digital environments (Primasari & Dwita, 2025). Moreover, the increasing digital intensity and adoption of e-commerce contribute not only to business performance but also to broader sustainability and economic growth, particularly within European Union countries (Criveanu, 2023).

Building on these developments, the growing role of social media and live-streaming platforms has also significantly shaped consumer behavior in e-commerce. Studies on e-commerce trends in Asia emphasize the integration of social media analytics and consumer engagement strategies to enhance online business performance (Aulawi et al., 2023). In addition, platform-specific research demonstrates how features of applications such as Douyin (TikTok) influence users' purchasing decisions, especially in live-streamed product promotions (Chen, 2022; Liu & Liang, 2025). These findings collectively suggest that integrating social media and interactive marketing strategies is essential for improving customer engagement and driving online sales.

Furthermore, the sustainability of e-commerce in developing economies and small and medium-sized enterprises (SMEs) is influenced by various barriers and enabling factors that affect long-term competitiveness. Studies conducted in Thailand highlight logistical, technological, and financial challenges faced by SMEs, emphasizing the need for strategies that improve operational efficiency and promote sustainable business practices (Amornkitvikai et al., 2022). These challenges, including limited access to advanced technologies and supply chain difficulties, can hinder the scalability of e-commerce solutions. In line with this, research in other developing countries underscores the importance of comprehensive strategic planning, integration of green economy initiatives, and sustainable digital growth approaches for successful e-commerce

implementation (Mustafa et al., 2022). In addition, recent studies emphasize the critical role of secure payment infrastructures and trust-building mechanisms in enhancing the performance of e-commerce systems. Hossain et al. (2024) highlight that online payment systems significantly affect customer trust and loyalty, particularly when users perceive strong security measures and convenient transaction processes. Supporting this, Handoyo (2024) explains that consumer behavior in digital marketplaces is shaped by multiple interrelated factors, including trust, perceived risk, system security, and electronic word-of-mouth (e-WOM). These findings indicate that improving payment security and reinforcing consumer confidence are essential components in sustaining user engagement and encouraging repeat online transactions in modern e-commerce environments.

Local Studies

The development of a web-based ordering and inventory management system is supported by studies emphasizing automation, efficiency, and database integration in business operations. Digital systems replace manual processes by automating product cataloging, order processing, payment handling, and record keeping, resulting in improved accuracy, productivity, and data-driven decision-making (Acharya, 2023). Centralized databases play a crucial role in storing customer and transaction data while enabling analysis of purchasing behavior, which supports effective pricing strategies and overall business performance (Mabeza, 2024). Moreover, modern e-commerce

architectures highlight the need for secure and scalable integration of databases and payment gateways to ensure safe and seamless transactions (Samonte et al., 2025).

These systems also enhance inventory management and forecasting by using customer data to predict demand, optimize stock levels, and prevent overstocking or shortages (Catubag et al., 2024). Integrating ordering, inventory, and database management into a unified platform improves operational efficiency, sales monitoring, and coordination among business functions (Bernasol et al., 2024). Additionally, web-based ordering systems improve service efficiency by reducing queues, minimizing errors, and increasing customer satisfaction (Babasa et al., 2025), while real-time inventory tracking ensures accurate order fulfillment and responsiveness (Barcelona et al., 2024).

Furthermore, the integration of secure electronic payment systems, such as GCash, is essential for modern platforms. Strong security measures—including encryption, authentication, and system safeguards—are necessary to prevent fraud and protect sensitive data (Tupas & Hernandez, 2024; Samonte et al., 2025). Incorporating these features enhances user trust, system reliability, and adoption, supporting the development of a secure and efficient web-based system with integrated ordering, inventory, and customer management functionalities.

Synthesis

The reviewed foreign and local literature and studies collectively emphasize that web-based ordering and inventory management systems are essential in improving operational efficiency, accuracy, and customer engagement in modern business environments. These systems integrate key functions such as inventory tracking, customer database management, order processing, and digital payments, enabling businesses to transition from manual to automated processes (Liew & Nayef, 2022; Acharya, 2023). Across both foreign and local contexts, consumer behavior, user experience, and loyalty programs are consistently identified as critical factors influencing customer satisfaction and repeat purchases, highlighting the importance of incorporating personalized features and reward mechanisms in e-commerce platforms (Panjaitan, 2021; Barrera et al., 2022; Primasari & Dwita, 2025). Furthermore, studies indicate that digital platforms, supported by data analytics and forecasting tools, enhance decision-making, optimize inventory levels, and contribute to overall business performance and sustainability (Catubag et al., 2024; Criveanu, 2023).

At the same time, both foreign and local studies highlight the growing importance of secure system architecture and digital payment integration in sustaining e-commerce operations. The adoption of cashless payment systems improves transaction efficiency and customer convenience but also introduces risks related to data privacy, fraud, and cybersecurity, necessitating the implementation of robust security measures such as encryption, authentication,

and secure payment gateways (Yao & Tarofder, 2024; Tupas & Hernandez, 2024; Samonte et al., 2025). Additionally, the integration of multiple system components—such as ordering, inventory, payment, and customer management—into a unified platform enhances reliability, reduces errors, and strengthens overall system performance (Bernasol et al., 2024). These findings collectively support the development of the study as a comprehensive solution that improves operational efficiency, ensures secure transactions, and enhances customer satisfaction and retention.

Chapter 3

RESEARCH METHODOLOGY

This chapter presents the research methodology and sources of information used by the researchers in the development of the proposed Web-Based Ordering and Inventory Management System with Customer Database and Loyalty Features. It includes the research design, population and sampling, respondents of the study, research instrument, and evaluation and scoring procedure. These methods were used to systematically gather data and evaluate the effectiveness, usability, and functionality of the developed system.

Research Design

The research approach for this study will be based on an Iterative Software Development Life Cycle (SDLC), which will be implemented using the Scrum framework. This approach enables the system to be developed in iterative cycles through structured sprint planning, development, testing, and review. It ensures that the system is implemented systematically while allowing continuous improvements based on feedback, in order to meet the specific objectives of the study, including improving ordering efficiency, monitoring inventory, and enhancing customer engagement through loyalty features.

System Development Method

A software development methodology is a structured process by which developers design, implement, and test new software systems. Following a methodology benefits developers because it provides a clear sequence of steps that guides professionals through each stage of development. Additionally, development methodologies often follow a design philosophy, which helps developers align their process and the system's features with its functional goals.

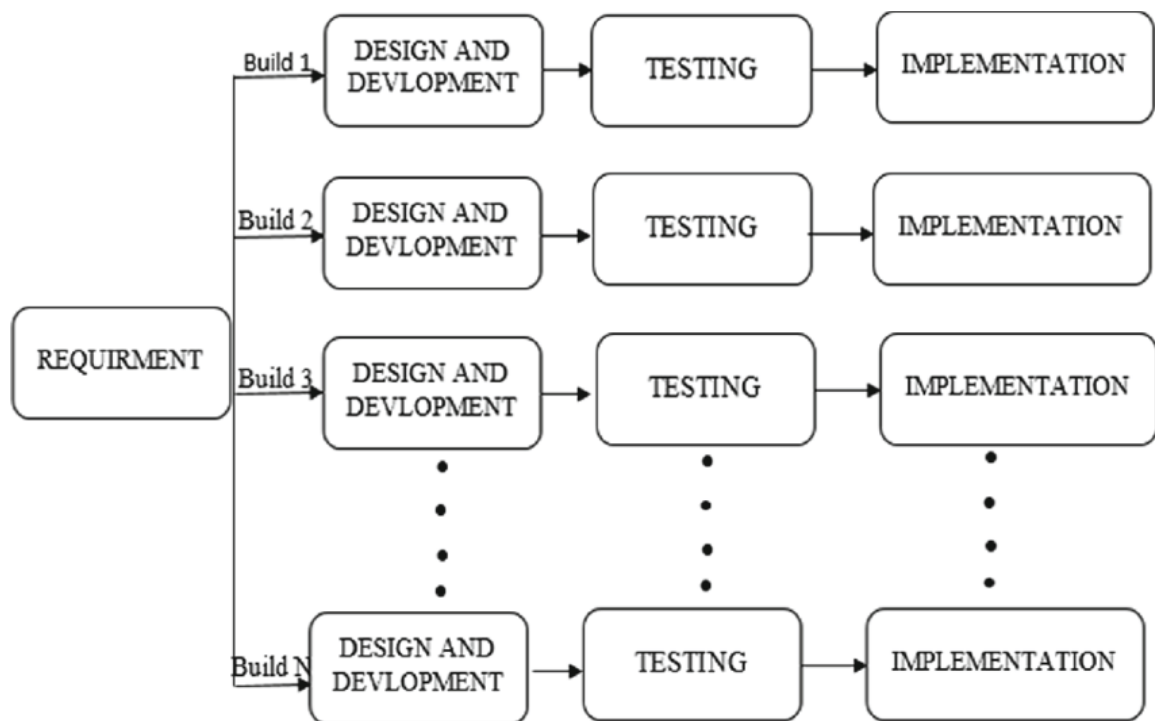


Figure 2. Iterative SDLC Workflow (Okesela, Adebiyi, & Owoade, 2020)

For this study, the researchers will adopt an Iterative Software Development Life Cycle (SDLC) as the primary development model. Within each

iteration, the Scrum framework will be applied to manage the development process through structured sprints, including sprint planning, development, testing, and review. This approach enables the system to be developed incrementally, allowing continuous feedback integration and progressive enhancement of system functionalities. It also ensures systematic delivery of features aligned with the study objectives, such as improving ordering efficiency, enhancing inventory monitoring, and strengthening customer engagement through loyalty features (Mishra & Alzoubi, 2023; Tetteh, 2024).

Development Cycle

The development cycle will follow the Iterative Software Development Life Cycle (SDLC), with Scrum sprints applied within each iteration to manage and organize development activities. Each iteration will begin with planning, where the goals, priorities, and requirements for that cycle are identified, followed by analysis of the functional specifications of the features to be developed. The design phase will then be conducted, involving the creation of wireframes, mockups, and system architecture to guide implementation. After design, the implementation phase will focus on developing and integrating system features into a working prototype, followed by testing to evaluate functionality, usability, and reliability. Finally, evaluation and feedback from stakeholders and potential users will be gathered and used as inputs for improvements in the next iteration. Through the application of Scrum within each iteration, the process ensures

continuous feedback integration, incremental development, and alignment of the system with the study's objectives.

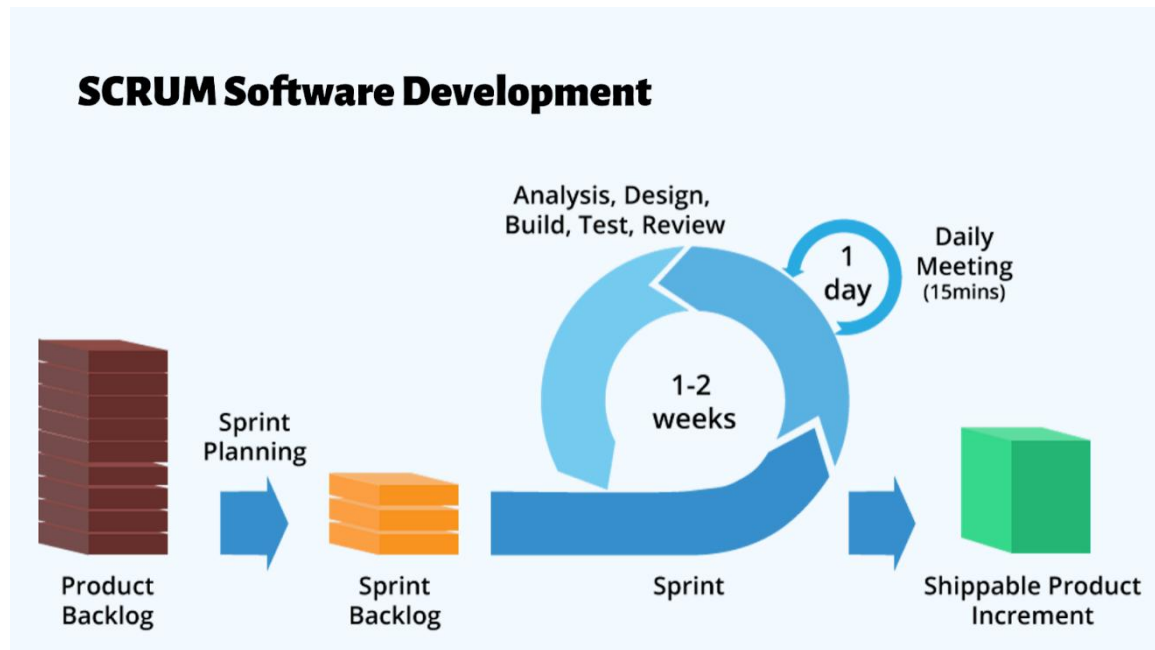


Figure 3. Scrum Workflow (Schwaber & Sutherland, 2020).

Input. The system requirements and selected technologies will serve as the foundation for the development of the proposed system. The requirements will define the expected functionalities, performance standards, and constraints, and will be analyzed to determine the core features to be implemented, including product catalog management, order processing, inventory monitoring, customer database management, and loyalty reward features. In addition, authentication and encryption mechanisms will be incorporated as essential requirements to ensure system security. Authentication will verify the identity of users through

secure login credentials, while encryption will protect sensitive data such as customer information, passwords, and transaction details during transmission and storage. Meanwhile, the selection of appropriate technologies will ensure compatibility with Iterative SDLC and Scrum methodologies, enabling efficient development, continuous testing, and seamless integration throughout each sprint.

Testing. The system will be conducted continuously throughout the Scrum workflow to ensure that each developed feature functions properly. After every sprint, system components such as customer registration and login, product management, order processing, inventory updates, and loyalty point computation will be tested for functionality, usability, and reliability. Additionally, security testing will be performed on authentication and encryption features, including verifying secure login processes, password protection, and proper data encryption during transmission and storage. Any errors or issues identified during testing will be recorded in the sprint review and addressed in the next sprint. This iterative testing approach allows the development team to improve system performance, strengthen data security, and ensure that all modules of the web-based ordering and inventory management system operate correctly.

Output. The primary output of the Scrum workflow will be a fully functional Web-Based Ordering and Inventory Management System with

Customer Database and Loyalty Features. Through iterative sprint cycles, the system will gradually develop key modules including product management, customer database management, online ordering, inventory monitoring, and loyalty points tracking. It will also include secure authentication and encrypted data handling, ensuring that user accounts, transactions, and sensitive information are protected from unauthorized access. Each completed sprint will produce a working component of the system that contributes to the final integrated platform. The completed system will allow administrators to manage products and inventory efficiently while enabling customers to place orders securely and accumulate loyalty rewards through the web-based platform.

Restrictive Conditions. Several restrictive conditions may affect the implementation of the Scrum workflow in the development of the system. The availability of users for testing and feedback may influence how effectively system features are evaluated during sprint reviews. Technical limitations such as internet connectivity, server performance, and compatibility of development tools may also impact development and testing activities. In addition, implementing authentication and encryption may require additional system resources, technical expertise, and careful configuration, which could pose challenges for the development team. Additionally, the skills and coordination of the development team may affect the efficiency of sprint planning, implementation, and review. By considering these conditions, the development team can improve collaboration, address issues promptly, and ensure the successful and secure completion of the system.

Population and Sampling

In this study, the developers will use purposive sampling. Purposive sampling is a form of non-probability sampling in which the developers will rely on their own judgment when selecting participants for the study. According to Frost (2022), purposive sampling is a strategy in which researchers intentionally select certain participants who can provide relevant insights and assist the study in fulfilling its objectives.

Table 1

Population and Distribution of Respondents

Category	Respondents	Percentage
Business Owners/Administrators	15	14.29%
Staff/Store Personnel	15	14.29%
IT/Computer Engineering Students	15	14.29%
Customers/Online Buyers	60	57.14%
Total	105	100%

The respondents will be chosen based on their experience with business operations, online ordering, inventory management, and customer interactions. These individuals will be considered essential for evaluating the proposed Web-Based Ordering and Inventory Management System with Customer Database and Loyalty Features. The subjects will be selected purposively because they can provide meaningful feedback on the system's functionality, usability, and overall effectiveness.

The population, sample size, and percentage distribution of responses will be presented in the table for each of the four groups: Business Owners/Administrators, Staff/Store Personnel, IT/Computer Engineering Students, and Customers/Online Buyers. The distribution of respondents will be as follows: Business Owners/Administrators, Staff/Store Personnel, and IT/Computer Engineering Students will each comprise 14.29 percent (15 out of 105 respondents), while Customers/Online Buyers will comprise the largest share at 57.14 percent (60 out of 105 respondents).

Respondents of the Study

The study will involve a total of one hundred five (105) respondents, consisting of fifteen (15) business owners/administrators, fifteen (15) staff/store personnel, fifteen (15) IT/Computer Engineering Students, and sixty (60) customers/online buyers. These individuals will participate in the testing phase of the system to evaluate its potential for application within the chosen business context.

The testing process will aim to assess the acceptability, usability, and functionality of the web-based ordering and inventory management system. Through surveys and feedback sessions, the researchers will gather valuable insights that will guide improvements to the system. The data and observations obtained from the respondents will contribute to the refinement and validation

of the design project, ensuring that it will meet the study's objectives and address the needs of its users.

Research Instrument

The research instruments that will be employed in this study will include a Likert scale questionnaire, consultations, system testing, and observations to be conducted before and after the implementation of the system. These instruments will be utilized to collect essential information, providing a comprehensive understanding of the proposed web-based ordering and inventory management system with a customer database and loyalty features.

The developers will prepare the questionnaires for approval and will seek permission from respondents prior to conducting the survey. The data sources of the study will primarily come from the initial survey and evaluations to be conducted during the final testing phase of the system. The instruments will be designed to collect feedback regarding the system's usability, functionality, efficiency, and overall user satisfaction.

In addition, related articles, theses, journals, and studies on e-commerce, web-based systems, inventory management, and customer loyalty programs will be reviewed to support the development of the system and its evaluation metrics. System testing and observations will provide practical insights into the system's

performance, particularly in terms of order processing, inventory tracking, and loyalty management functionalities. Furthermore, essential security measures, including user authentication and data protection, will be incorporated during system testing to ensure the confidentiality of user information.

After respondents complete the surveys and test the system, necessary modifications and improvements will be implemented to ensure that the system and the instruments are valid, reliable, and aligned with the study objectives. The combination of questionnaires, consultations, observations, and testing will enable the developers to gather both quantitative and qualitative data, ensuring a thorough evaluation of the system's effectiveness and suitability for its intended users.

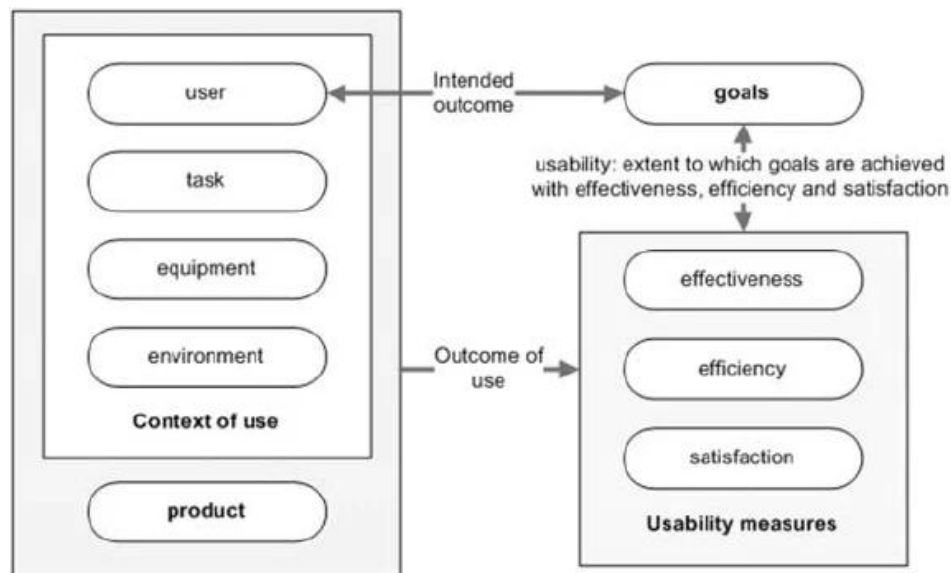


Figure 4. ISO 9241-11 Usability Evaluation Framework (International Organization for Standardization, 2018).

Figure 4 will illustrate the Usability Evaluation Framework based on ISO 9241-11, which measures effectiveness, efficiency, and user satisfaction. The developers will apply this framework to assess the proposed web-based ordering and inventory management system with customer database and loyalty features. Data will be collected through questionnaires, observations, and system testing to evaluate task accuracy, completion time, ease of navigation, and overall user experience. Feedback will be used to improve the interface design, functionality, and usability of the system, ensuring it meets the needs of both staff and customers, while maintaining secure access through user authentication and data protection measures.

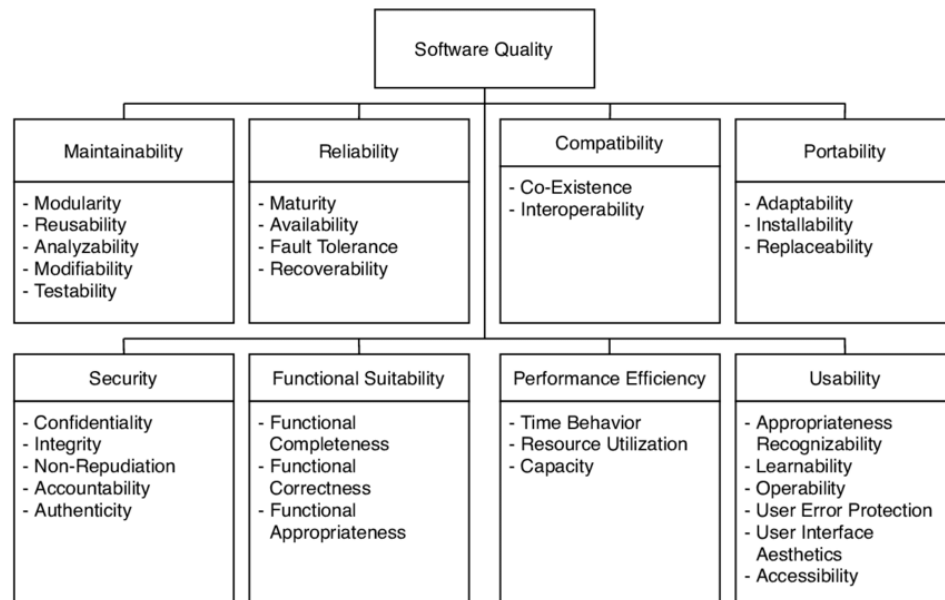


Figure 5. ISO/IEC 25010 Software Quality Model (Consortium for Information & Software Quality, 2022)

Figure 5 will depict the ISO/IEC 25010 Software Quality Model, which assesses software quality across eight characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. This model will serve as a standard reference for evaluating the technical quality of the system. After respondents complete system testing and surveys, the developers will use this model to guide improvements and ensure that the system meets recognized software quality standards.

Evaluation and Scoring

Likert scales are widely used in survey research to measure respondents' attitudes, perceptions, and levels of agreement toward specific statements. These scales are commonly applied in structured questionnaires that contain predefined response options, allowing researchers to quantify opinions and perceptions for statistical analysis (Jebb et al., 2021). In this study, the developers will use a Likert scale questionnaire to evaluate the acceptability, effectiveness, and reliability of the proposed Web-Based Ordering and Inventory Management System with Customer Database and Loyalty Features. The evaluation will be conducted after respondents interact with the developed system and assess its functionalities, including online ordering, inventory monitoring, customer database management, and loyalty reward features, as well as their perception of basic security measures such as user authentication and

data protection. The responses gathered from the survey questionnaire will then be analyzed to determine the overall performance and usability of the system.

Table 2

Likert Scale of the Level of Acceptability of the System

SCALE	MEAN RANGE	VERBAL INTEPRETATION
4	3.25 – 4.00	Strongly Acceptable
3	2.50 – 3.24	Acceptable
2	1.75 – 2.49	Unacceptable
1	1.0 – 1.74	Strongly Unacceptable

Table 2 will present the Likert Scale used to evaluate the level of acceptability of the system. A mean value ranging from 1.00 to 1.74 will indicate a strongly unacceptable interpretation, while 1.75 to 2.49 will indicate that the system is unacceptable. A mean value ranging from 2.50 to 3.24 will suggest that the system is acceptable, and 3.25 to 4.00 will indicate that the system is strongly acceptable.

The scale will be used by the developers to determine the overall acceptability of the proposed web-based ordering and inventory management system in terms of usability, functionality, and user satisfaction.

Table 3

Scale of the Level of Effectiveness of the System

SCALE	MEAN RANGE	VERBAL INTEPRETATION
4	3.25 – 4.00	Very Effective
3	2.50 – 3.24	Effective
2	1.75 – 2.49	Ineffective
1	1.0 – 1.74	Very Ineffective

Table 3 will show the Likert scale used to measure the effectiveness of the proposed system. A mean value ranging from 3.25 to 4.00 will indicate that the system is very effective, while 2.50 to 3.24 will indicate that the system is effective. A mean value between 1.75 and 2.49 will indicate that the system is ineffective, while 1.00 to 1.74 will indicate that the system is very ineffective.

This scale will be used to evaluate how well the system performs its intended functions such as order processing, inventory monitoring, customer database management, and loyalty reward features.

Table 4

Scale of the Level of Reliability of the System

SCALE	MEAN RANGE	VERBAL INTEPRETATION
5	91% and above	Very Reliable
4	81% – 90%	Reliable
3	71% – 80%	Fairly Reliable
2	61% – 70%	Unreliable
1	60% and below	Very Unreliable

Table 4 will present the 5-point Likert scale used to measure the reliability of the system. A score of 1 with a percentage value of 60% or below will indicate that the system is very unreliable. A score of 2 with a percentage range of 61% to 70% will indicate that the system is unreliable. A score of 3 with a percentage range of 71% to 80% will suggest that the system is fairly reliable. A score of 4 with a percentage range of 81% to 90% will indicate that the system is reliable, while a score of 5 with a percentage value of 91% and above will indicate that the system is very reliable.

The reliability scale will be used by the developers to determine whether the proposed system can consistently perform tasks such as processing orders, updating inventory records, storing customer information, and managing loyalty points without system failure or errors.

Data Gathering Procedure

The developers will collect the necessary data through consultations, system testing, observations, and survey questionnaires. Prior to the evaluation, permission will be obtained from the selected respondents, which will include business owners/administrators, store personnel, IT/Computer Engineering students, and customers or online buyers. These respondents will be asked to evaluate the proposed Web-Based Ordering and Inventory Management System with Customer Database and Loyalty Features and provide feedback regarding its usability, functionality, and overall performance.

The questionnaire will consist of standardized closed-ended questions arranged in a structured format, allowing respondents to select answers from predefined response options using a Likert scale. Structured questionnaires are widely used in research because they allow consistent data collection and easier analysis of respondents' perceptions and experiences (Jebb et al., 2021).

The respondents will be given time to interact with the system by browsing products, placing orders, monitoring inventory updates, and exploring the loyalty features once the system is developed. During the testing phase, the developers will observe how users interact with the system to identify possible technical issues, errors, or usability concerns.

The feedback, observations, and suggestions gathered from the respondents will be carefully analyzed and used to improve the system and ensure that it effectively meets the needs of its intended users.

Statistical Treatment of Data

The developers will analyze and interpret the collected data using the following statistical tools in order to evaluate the performance of the proposed system:

1. **Weighted Mean.** This statistical tool will be used to compute and determine the level of acceptability, effectiveness, and reliability of the proposed Web-Based Ordering and Inventory Management System with Customer Database and Loyalty Features. The weighted mean will be applied to calculate the average responses of the respondents based on the Likert scale questionnaire.
2. **Composite Mean or Average.** This will be used to determine and validate the overall mean values of the system's acceptability, effectiveness, and reliability. The composite mean will represent the general average of all computed weighted means based on the responses of the respondents.
3. **Percentage.** This statistical tool will be used to determine the proportion of responses obtained from the respondents. It will help present the distribution of responses and identify the frequency of each response category in the evaluation of the system.

System Design

The system design presents the overall structure and operation. It includes system flowcharts, process diagrams, and architectural diagrams that illustrate how the system functions. The flowchart diagram shows the step-by-step process of how users interact with the website, including product browsing,

order placement, inventory updates, and loyalty point management. The block diagram illustrates the relationship between the main components of the system, such as the user interface, database, ordering module, inventory module, and loyalty feature module. The data flow diagram shows the flow of information between users, system processes, and the database. The diagram identifies the main processes such as user registration and login, product browsing, order placement, inventory updating, and loyalty point management. It also shows how data is stored and retrieved from the database, including customer information, product details, order records, and loyalty points. In addition, the system design incorporates security mechanisms such as user authentication during login and data encryption to protect sensitive information stored and transmitted within the system.

System Block Diagram

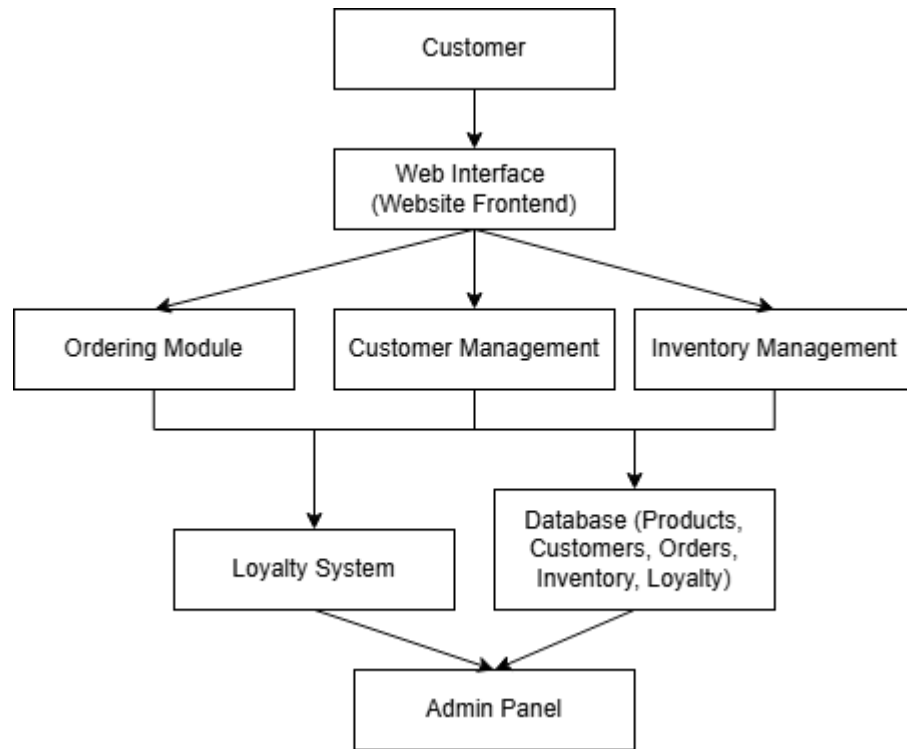


Figure 6. System Block Diagram.

The block diagram of the Web-Based Ordering and Inventory Management System with Customer Database and Loyalty Features illustrates the main components of the system and how they interact with one another. The customer represents the users who access the website to browse available products, place orders, and earn loyalty points from their purchases. These users interact with the system through the web interface, which serves as the front-end of the website where customers can log in, view products, and complete transactions. The ordering module processes customer orders and records transaction details. Meanwhile, the customer management component handles user registration and

login authentication, ensuring that only verified users can access their accounts and system features. The inventory management module monitors and updates product stock levels to ensure accurate product availability within the system. The loyalty system calculates and records the loyalty points earned by customers based on their purchases, encouraging repeat transactions. All information processed by the system is stored in the database, which contains records of products, customers, orders, inventory levels, and loyalty points, with sensitive data being protected through encryption to ensure data confidentiality and security. Lastly, the Admin Panel allows administrators to manage system operations such as updating product information, monitoring inventory, reviewing orders, and maintaining customer records.

System Flowchart Diagram

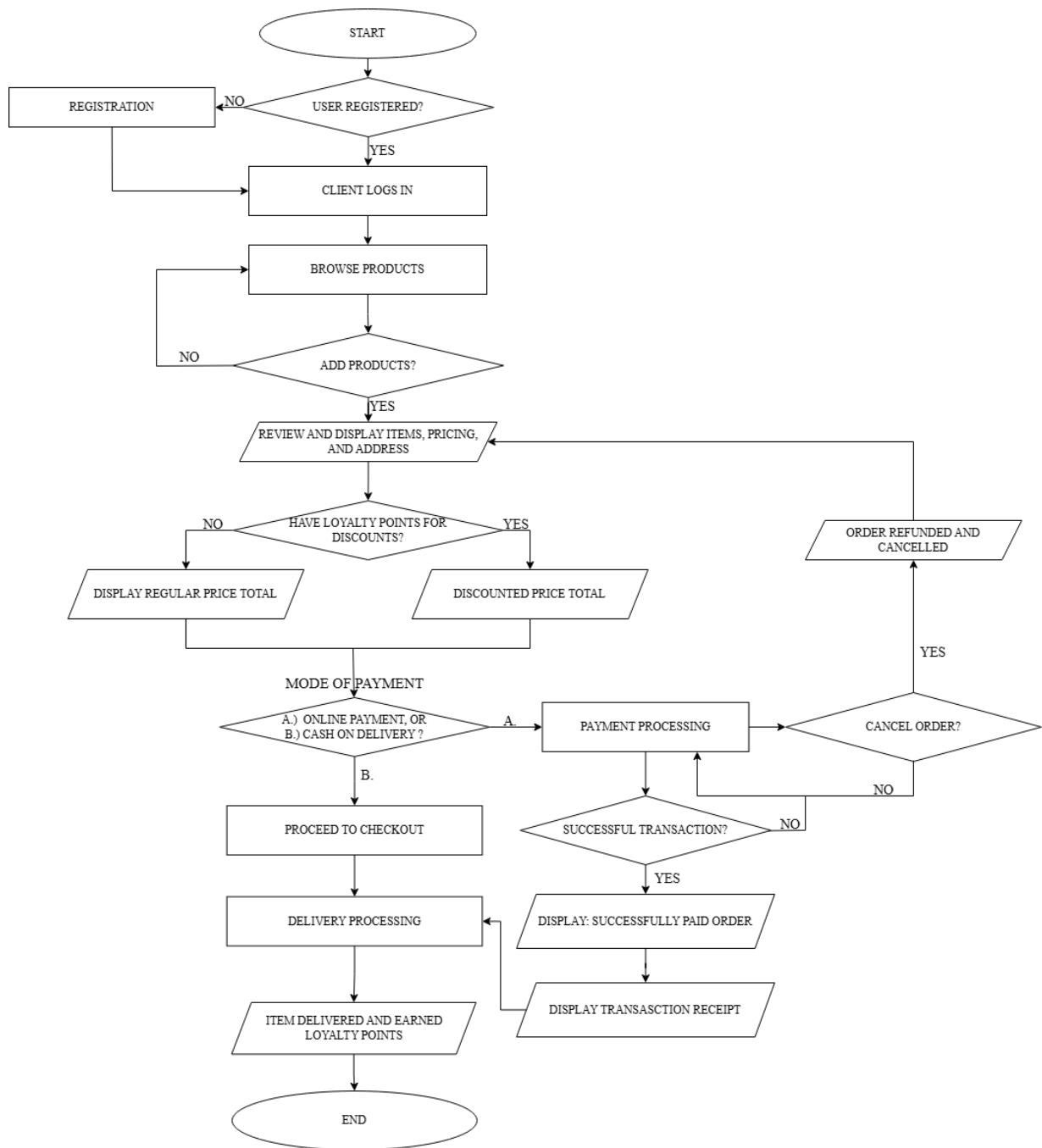


Figure 7. System Flowchart Diagram.

The flowchart illustrates the overall process of a web-based ordering and inventory management system with integrated customer interaction and loyalty features. It begins with user registration and login, followed by product browsing and item selection. Once items are added, the system displays order details including pricing and delivery address, while also checking whether the customer has accumulated loyalty points based on purchase frequency. The system incorporates a reward mechanism wherein customers receive incentives after a defined number of successful transactions, such as every 3rd, 5th, or 10th purchase. For example, if a customer completes five successful orders, corresponding loyalty points are automatically added to their account. Depending on the availability of points, either a regular or discounted total is computed before proceeding to the payment stage.

The process then branches into payment options, including online payment or cash on delivery, with corresponding steps for payment processing, transaction validation, and possible order cancellation or refund. A successful transaction leads to checkout and delivery processing, culminating in the confirmation of a completed order. Upon delivery, the system updates the customer's purchase count and awards loyalty points accordingly, reinforcing repeat engagement and sustained system use. Overall, the flowchart demonstrates a systematic and user-centered approach that integrates ordering, payment, inventory tracking, and purchase-based loyalty mechanisms within a single platform.

System Data Flow Diagram

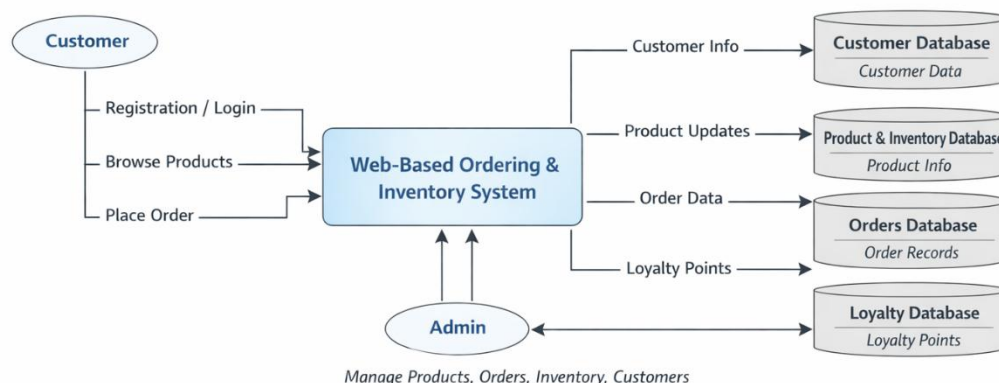


Figure 8. System Data Flow Diagram.

The data flow diagram (DFD) illustrates how information moves within the Web-Based Ordering and Inventory Management System with Customer Database and Loyalty Features. In this system, the Customer provides inputs such as registration details, login credentials, product orders, and requests to view available products. These inputs are processed by the Web-Based Ordering and Inventory System, where user login credentials undergo authentication to verify access before allowing interaction with system features such as ordering and product browsing. The system manages customer actions, processes orders, updates product inventory, and calculates loyalty points earned from purchases. It stores and retrieves customer information from the Customer Database, which contains user profiles and account details. Product information and stock levels are maintained in the Product and Inventory Database, ensuring accurate monitoring of available items. All completed transactions and order details are

recorded in the Orders Database for tracking and reference. Meanwhile, the Loyalty Database stores and updates the loyalty points earned by customers based on their purchase activities. In addition, the admin interacts with the system to manage products, monitor inventory levels, review customer orders, and maintain customer records, ensuring that the system operates efficiently and accurately.

System Development

The development involved the use of various software tools, programming languages, and development frameworks to ensure the system operates efficiently and meets the requirements of both customers and administrators are listed below:

System Software Development



Figure 9. *Typescript (Williams, 2023).*

TypeScript is an open-source programming language developed by Microsoft that extends JavaScript by introducing static typing and additional features to improve code structure, reliability, and scalability. It is considered a strict syntactical superset of JavaScript, which means that all valid JavaScript code is also valid TypeScript code. Through its support for explicit type definitions and compile-time error checking, TypeScript enables developers to detect potential issues early in the development process, resulting in more maintainable and robust applications. Furthermore, TypeScript code is transpiled into standard JavaScript, allowing it to run seamlessly across various web browsers and development environments while maintaining compatibility with existing JavaScript libraries and frameworks (Williams, 2023). It is a statically-typed superset of JavaScript developed by Microsoft that enforces strict type checking during the development process. By defining data types explicitly, it significantly reduces the likelihood of runtime errors and improves overall code maintainability, making it well-suited as the primary language for building scalable full-stack applications.



Figure 10. *Structure Query Language (Jonker, 2025).*

SQL stands for Structured Query Language and is the standard, domain-specific programming language used for managing and accessing data stored in relational database management systems (RDBMS) (e.g., MySQL, Oracle, Microsoft SQL Server, PostgreSQL). (Jonker, 2025). It is designed to work with structured data organized into tables consisting of rows and columns. It is the standardized language used for managing and manipulating relational databases. It enables developers to perform essential database operations such as data retrieval, insertion, modification, and deletion, serving as the primary means of communication between the application and its underlying database system.



Figure 11. *JavaScript (ECMA International, 2020)*

JavaScript is a high-level, often just-in-time-compiled language that conforms to the ECMAScript standard. It has dynamic typing, prototype-based object-orientation, and first-class functions. It is multi-paradigm, supporting event-driven, functional, and imperative programming styles. It has application programming interfaces (APIs) for working with text, dates, regular expressions, standard data structures, and the Document Object Model (DOM) (ECMA International, 2020). It is the foundational scripting language of modern web development. While it is not directly written in this system due to the use of TypeScript, it remains the runtime language that powers all client-side and server-side execution, as TypeScript ultimately compiles down to JavaScript before being executed.

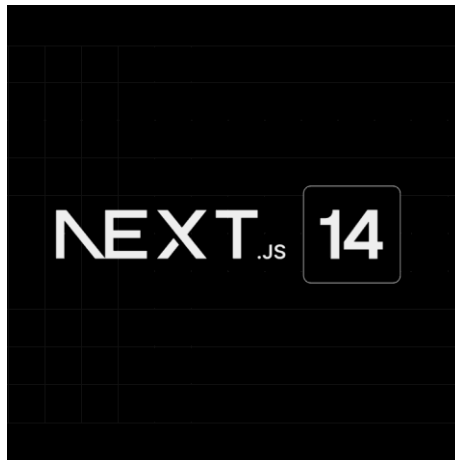


Figure 12. *Next.js Version 14+ (Robinson, 2023).*

Next.js v14+ is a modern React-based web development framework that enhances the creation of high-performance and scalable web applications. It introduces significant improvements in stability, speed, and developer experience, particularly through the integration of Server Actions, optimized build processes, and advanced rendering capabilities. This version emphasizes efficient full-stack development by enabling seamless interaction between frontend and backend components within a unified framework (Robinson, 2023). It is a full-stack React-based web framework that provides a robust foundation for building modern web applications. It supports multiple rendering strategies including server-side rendering, static site generation, and client-side rendering, allowing the system to optimize performance and user experience depending on the nature of each page or feature.



Figure 13. *Supabase (Supabase, 2020).*

Supabase is an open-source backend platform that serves as an alternative to Firebase, providing developers with a complete backend infrastructure built on PostgreSQL. It enables relational data management using SQL and includes integrated services such as authentication, file storage, real-time data subscriptions, and serverless edge functions within a unified system. By combining these features into a single platform, Supabase allows developers to efficiently build and scale modern applications while leveraging the flexibility, security, and transparency of open-source technologies (Pangea, 2026). It is an open-source backend-as-a-service platform built on top of PostgreSQL. It provides a comprehensive suite of backend tools including a managed relational database, authentication services, real-time subscriptions, and an auto-generated RESTful API layer through PostgREST, significantly reducing the need for manual backend infrastructure setup.

System Architecture

The system architecture of this study presents the overall structure and organization of the software components used. As a purely software-based solution, the architecture integrates the front-end interface, back-end services, and database into a unified and accessible platform. The system follows a structured web architecture in which users interact with the application through a browser-based interface, while the back-end handles business logic, data processing, and system operations. The database serves as the central repository for storing product information, customer records, transaction details, and loyalty data.

System Software Architecture

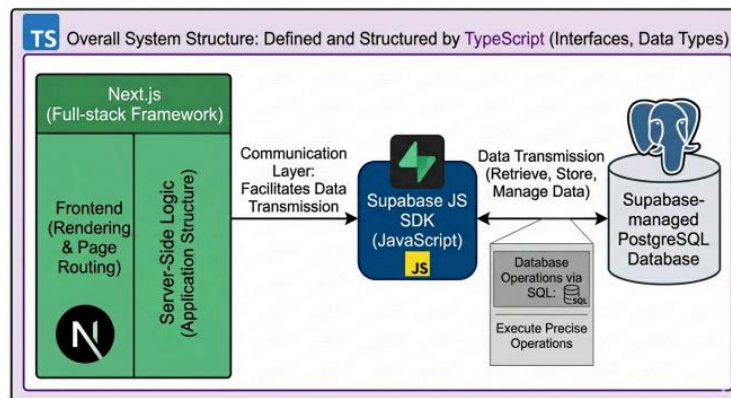


Figure 14. System Software Integration.

TypeScript serves as the primary language that structures and defines the logic of the entire system, ensuring that all components interact with clearly

defined data types and interfaces. These components are organized and rendered through Next.js, which acts as the full-stack framework responsible for handling page routing, server-side logic, and the overall application structure. Through the Supabase JS SDK, which is built on JavaScript, Next.js establishes a communication layer that facilitates the transmission of data between the frontend and the Supabase-managed PostgreSQL database. SQL is then utilized within this layer to execute precise database operations, enabling the system to efficiently retrieve, store, and manage data as required by the application.

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APPENDICES

Appendix A

Letter of Intent for Conducting Survey

February 20, 2026

TRIMEX COLLEGES, INC.

Computer Engineering Department

Trojan Building, Brgy Poblacion, City of Biñan, Laguna

Subject: Request for Permission to Conduct Research Survey and Evaluation

Dear Sir/Ma'am,

Greetings!

We are Computer Engineering students from Trimex Colleges, currently conducting a research study titled **“WEB-BASED ORDERING AND INVENTORY MANAGEMENT SYSTEM WITH CUSTOMER DATABASE AND LOYALTY FEATURES”** as part of our thesis requirement.

The objective of our study is to develop and evaluate a web-based system that streamlines ordering and inventory management, integrates a customer database, and incorporates loyalty features to enhance user experience and operational efficiency.

In line with this, we would like to request your permission to conduct our research within your establishment and to identify **one hundred five (105) potential participants**, including staff and users, who can provide insights through interaction with our system.

The research involves filling out questionnaires, testing and interacting with our web-based system, survey participation, and evaluation. Rest assured that all data gathered will be treated with strict confidentiality and will be used solely for academic purposes.

We look forward to your favorable response. Should you have any questions or clarifications, you may contact us at mauirodriguez@trimexcolleges.edu.ph or 0991-884-4994.

We sincerely hope for your approval and support for this research study.

Thank you very much.

Sincerely,

Student Researchers,

BS Computer Engineering Students,

Trimex Colleges, Inc.

Noted by:

Appendix B

Survey Questionnaire and System Quality Evaluation Form

SURVEY QUESTIONNAIRE AND SYSTEM QUALITY EVALUATION FORM

Dear Respondents,

Greetings!

We are currently third-year students in the Bachelor of Science in Computer Engineering (BSCpE) of Trimex Colleges this Academic Year of 2026-2027. As part of our requirement for the course of Software Engineering (SoftEng), we are conducting a system development called ***“A WEB-BASED ORDERING AND INVENTORY MANAGEMENT SYSTEM WITH DATABASE CUSTOMERS WITH LOYALTY FEATURES”***. This study aims to develop and evaluate a website that facilitates online ordering, inventory monitoring, and customer database management while incorporating a loyalty reward system to improve customer engagement and business operations. The purpose of this survey is to evaluate the functionality, efficiency, reliability, and overall quality of the developed system. your responses will help determine whether the system effectively supports business processes such as order processing, inventory tracking, and customer management.

In light of this, we would like to request your cooperation to participate in answering a survey questionnaire and system quality evaluation. Rest assured that the data and information provided will be kept confidential.

Thank you for your time and effort.

Regards,



Alexandra Bascugin



Maui Rodriguez

REPUBLIC ACT NO. 10173 (DATA PRIVACY ACT)

Thank you for your willingness to participate in this survey. We value your privacy and want to assure you that your information will be handled responsibly in accordance with Data Privacy Act, Republic Act No. 10173 for the Philippines.

I. INFORMATION WE COLLECT

This survey may collect some or all of the following types of information from you:

PERSONAL DATA: This includes information that can be used to directly or indirectly identify you, including your name, email address, age, sex, and other demographic information.

NON-PERSONAL DATA: This includes information that cannot be used to identify you, such as your opinions, preferences, and responses to survey questions.

II. HOW WE USE YOUR INFORMATION

The information you provide will be used for the following purposes:

- To analyze and interpret the results of the survey
- To improve our system development
- To understand the needs of the guests

III. YOUR RIGHTS AS A RESPONDENT

You have the following rights under the Data Privacy Act:

- **RIGHT TO BE INFORMED:** You have the right to be informed about the collection, use, and disclosure of your personal data. This agreement provides that information.
- **RIGHT TO ACCESS:** You have the right to access your personal data and request a copy of it.
- **RIGHT TO RECTIFICATION:** You have the right to have your personal data corrected if it is inaccurate or incomplete.
- **RIGHT TO ERASURE OR BLOCKING:** You have the right to request that your personal data be erased or blocked from further processing.
- **RIGHT TO OBJECT:** You have the right to object to the processing of your personal data.
- **RIGHT TO DATA PORTABILITY:** You have the right to request that your personal data be transferred to another data controller.

IV. DATA SECURITY

We will take all reasonable steps to protect your personal data from unauthorized access, disclosure, alteration, or destruction.

V. DATA RETENTION

We will retain your personal data for as long as necessary to achieve the purposes for which it was collected, or as required by law.

VI. CONTACT US

If you have any questions regarding this Data Privacy Agreement, you may raise your concerns at any time by contacting us at the following email address:

mauirodriguez@student.trimexcolleges.edu.ph

alexandralouizelbascugin@student.trimexcolleges.edu.ph

Likert Scale Guide

Likert Scale for Acceptability & Effectiveness

1 – Strongly Disagree / Very Ineffective

2 – Disagree / Ineffective

3 – Agree / Effective

4 – Strongly Agree / Very Effective

Likert Scale for Reliability

1 – Very Unreliable

2 – Unreliable

3 – Fairly Reliable

4 – Reliable

5 – Very Reliable

Part I. Demographic Information

Age: _____

Gender:

☐ Male

☐ Female

☐ Prefer not to say

Role / Position:

☐ Business Owner / Administrator

☐ Staff / Store Personnel

☐ IT / Computer Engineering Student

☐ Customer / Online Buyer

Years of Experience in Business Operations / Online Ordering:

Part II. System Evaluation**FUNCTIONAL SUITABILITY****FUNCTIONAL APPROPRIATENESS**

The system provides functions that are appropriate and effective in supporting user tasks such as ordering, inventory management, and customer account operations.

Mark only one oval.

- ☐ Strongly Acceptable
- ☐ Acceptable
- ☐ Not Acceptable
- ☐ Strongly Unacceptable

FUNCTIONAL COMPLETENESS

The system provides all necessary functions required to fully support user requirements such as ordering, payment processing, inventory tracking, and loyalty management.

Mark only one oval.

- ☐ Strongly Acceptable
- ☐ Acceptable
- ☐ Not Acceptable
- ☐ Strongly Unacceptable

FUNCTIONAL CORRECTNESS

The system produces accurate outputs and performs functions correctly, including order processing, inventory updates, and transaction handling.

Mark only one oval.

- ☐ Strongly Acceptable
- ☐ Acceptable
- ☐ Not Acceptable
- ☐ Strongly Unacceptable

PERFORMANCE EFFICIENCY

The system represents its functions within specified time and throughput parameters and is efficient in the use of resources.

TIME BEHAVIOR

The system responds within acceptable time limits while performing tasks such as loading products, processing orders, and updating records efficiently.

Mark only one oval.

- ☐ Strongly Acceptable
- ☐ Acceptable
- ☐ Not Acceptable
- ☐ Strongly Unacceptable

Part III. Acceptability of the System

Indicate how much you agree with the following statements.

The system interface is user-friendly and easy to navigate.

- ☐ Strongly Agree
- ☐ Agree
- ☐ Disagree
- ☐ Strongly Disagree

The process of placing an order online is simple and convenient.

- ☐ Strongly Agree
- ☐ Agree
- ☐ Disagree
- ☐ Strongly Disagree

Inventory monitoring features are easy to understand and use.

- ☐ Strongly Agree
- ☐ Agree
- ☐ Disagree
- ☐ Strongly Disagree

Managing customer profiles and loyalty points is straightforward.

- Strongly Agree
- Agree
- Disagree
- Strongly Disagree

Overall, I find the system suitable for business operations and customer interaction.

- Strongly Agree
- Agree
- Disagree
- Strongly Disagree

Part IV. Effectiveness of the System

Processing online orders accurately and promptly.

- Very Effective
- Effective
- Ineffective
- Very Ineffective

Updating inventory in real time without errors.

- Very Effective
- Effective
- Ineffective
- Very Ineffective

Maintaining accurate customer records.

- Very Effective
- Effective
- Ineffective
- Very Ineffective

Calculating and tracking loyalty points correctly.

- Very Effective
- Effective
- Ineffective
- Very Ineffective

Supporting business operations efficiently and improving workflow.

- Very Effective
- Effective
- Ineffective
- Very Ineffective

Part V. Reliability of the System

The system consistently processes orders without failure.

- Very Reliable
- Reliable
- Fairly Reliable
- Unreliable
- Very Unreliable

Inventory updates and stock monitoring are performed accurately.

- Very Reliable
- Reliable
- Fairly Reliable
- Unreliable
- Very Unreliable

Customer database management is stable and secure.

- Very Reliable
- Reliable
- Fairly Reliable
- Unreliable
- Very Unreliable

Loyalty points are reliably calculated and recorded.

- ☐ Very Reliable
- ☐ Reliable
- ☐ Fairly Reliable
- ☐ Unreliable
- ☐ Very Unreliable

Overall, the system operates consistently without technical issues.

- ☐ Very Reliable
- ☐ Reliable
- ☐ Fairly Reliable
- ☐ Unreliable
- ☐ Very Unreliable

Part VI: Additional Feedback (Optional)

1. What features of the system did you find most helpful?

2. What improvements would you suggest for the system?

3. Additional comments or observations:

Thank you once again for your kindness and cooperation. We truly appreciate your help in making our academic journey more meaningful.

If you have any questions or concerns, please do not hesitate to reach out.

Appendix C**Individual Progress Report**

This form should be filled up and signed by the Thesis/Project Study/Feasibility Study/Design Project Adviser(s) every consultation of the researcher(s) in person or by e-mail.

Research Title:

**A WEB-BASED ORDERING AND INVENTORY MANAGEMENT SYSTEM WITH
DATABASE CUSTOMERS WITH LOYALTY FEATURES**

Researcher: **Alexandra Louizel F. Bascugin**

Contributions:

- **Manuscript Writing**
- **Manuscript Revising**
- **System Development**

Remarks:

*Follows instructions while demonstrating initiative during the completion of the system.

*Note: *Peer Evaluation*

Certified and validated by:

Dr. Kierven De Mesa

Dean, College of Engineering

This form should be filled up and signed by the Thesis/Project Study/Feasibility Study/Design Project Adviser(s) every consultation of the researcher(s) in person or by e-mail.

Research Title:

**A WEB-BASED ORDERING AND INVENTORY MANAGEMENT SYSTEM WITH
DATABASE CUSTOMERS WITH LOYALTY FEATURES**

Researcher: **Maui Rodriguez**

Contributions:

- **Manuscript Writing**
- **Manuscript Revising**
- **System Development**

Remarks:

*Demonstrating leadership and great communication skills towards members.

*Note: *Peer Evaluation*

Certified and validated by:

Dr. Kierven De Mesa

Dean, College of Engineering

Appendix D
Curriculum Vitae

BASCUGIN, ALEXANDRA LOUIZEL F.



CONTACT INFORMATION:

Address: #6 Pacita Ave., Brgy. Pacita II, City of San Pedro, Laguna

Mobile No.: (+63) 9942194241

E-mail Address: alexandrabascugin057@gmail.com

PERSONAL INFORMATION:

Birth: August 09, 2005

Age: 20

Physical Attributes: 43kg and 147.32 cm

Language: Tagalog and English

OBJECTIVE:

To obtain a position where academic knowledge and technical skills can be applied in a professional environment while gaining valuable experience, contributing to organizational goals, and continuously developing personal and professional competencies.

EDUCATIONAL BACKGROUND:

College:

Trimex Colleges

Bachelor of Science in Computer Engineering

Brgy. Poblacion, City of Biñan, Laguna, 4024

*(2024 - Present)***Senior High School:**

Immaculate Heart of Mary School

Science, Technology, Engineering, and Mathematics

Avocado St., Pacita II Phase II, City of San Pedro,
Laguna, 4023

(2021 - 2023)

Junior High School:

Immaculate Heart of Mary School

Avocado St., Pacita II Phase II, City of San Pedro,
Laguna, 4023

(2017 - 2021)

Elementary:

Holy Children Academy

32 San Cristobal St., Phase 6 Pacita Complex I, City
of San Pedro, Laguna, 4023

(2011 - 2017)

PROFESSIONAL SKILLS:

- Knowledgeable in using Microsoft Office (Word, Excel, PowerPoint)
- Knowledgeable in Programming (Python, Java)
- Average knowledge in HTML, CSS, JS (front-end to back-end programming)
- Basic knowledge in using programming software (Visual Studio Code, Eclipse, NetBeans)
- Proficient in computer networking (Remote, Wiring)
- Knowledgeable in hardware network systems (Routers, Switches)

- Adept in multimedia projects (Adobe Photoshop, Canva, Capcut)
- Strong writing and creativity skills

RODRIGUEZ, MAUI



CONTACT INFORMATION:

Address: Brgy. San Lorenzo Ruiz, City of San Pedro, Laguna

Mobile No.: (+63) 9918844994

E-mail Address: mauirodriguez0278@gmail.com

PERSONAL INFORMATION:

Birth: August 27, 2002

Age: 23

Physical Attributes: 50kg and 154.94 cm

Language: Filipino and English

OBJECTIVE:

To obtain a position that aligns with my knowledge and abilities, allowing me to further develop and maximize my professional skills. Additionally, to apply and integrate real-world experience and knowledge that can significantly contribute to the organization.

EDUCATIONAL BACKGROUND:

College:

Trimex Colleges

Bachelor of Science in Computer Engineering

Brgy. Poblacion, City of Biñan, Laguna

*(2024 - Present)***Senior High School:**

San Pedro College of Business Administration

KM30, Old National Highway, Brgy. Nueva, City of
San Pedro, Laguna, 4023*(2020 - 2022)***Junior High School:**

NBL Education System

Mindanao Drive Stage 5 Macaria Village, City of
Biñan, Philippines, 4024

(2016 - 2020)

Elementary:

FBC - Binan Christian School

B7 L14 St. Francis V Canlalay Biñan City Laguna,
City of Binān, Philippines, 4024

(2009 - 2016)

PROFESSIONAL SKILLS:

- Knowledgeable in using Microsoft Office (Word, Excel, PowerPoint).
- Knowledgeable in Python and C++.
- Intermediate knowledge in HTML, CSS, and JavaScript (front-end development).
- Basic ability to handle JSON data and parse responses in JavaScript.
- Familiarity with CSS frameworks (Bootstrap, Tailwind CSS).
- Basic testing using Jasmine for unit testing and code validation.
- Basic knowledge of hardware systems, including Arduino, ESP32, and other microcontrollers.
- Proficient skills in multimedia (video production, photography, and editing).

- Good communication and creativity skills.
- Strong leadership and writing skills.
- Intermediate knowledge in computer network troubleshooting and configuration.